Master Marine Physics 1st year Geophysical Fluid Dynamics

Problem sheet # 1 : Geophysical fluid kinematics and dynamics, Coriolis force, buoyancy force

1 Streamlines and trajectories

A vortex, with angular velocity ω drifts along the X axis with uniform velocity U. Compute the particle trajectories, the streamlines, with the references $x = R, y = 0$, $u = U, v = R\omega$ at $t = 0$.

2 Lagrangian and Eulerian derivatives

A ship cruises northward at $10km/h$. The surface pressure increases northwestward by $5Pa/km$. Which rate of pressure variation will an observer measure on a nearby island ? The pressure measured onboard the ship varies by $100Pa/3h$.

3 Hydrostatic pressure

Compare the decrease of pressure with altitude in the Boussinesq approximation (constant density) and for an isothermal atmosphere (law of ideal gas $p = \rho RT$).

4 Centrifugal force

A train drives at uniform speed $50km/h$ in a uniform curve. The rails are inclined with respect to the horizontal plane. The passengers feel an apparent increase of gravity by 10%. What is the radius of curvature of this curve ?

5 The Wilhelm Reich experiment

A stone falls from a bridge 330 m high. Compute its N-S and E-W deflections due to both components of the Coriolis force (assume that the latitude

is 45°N).

6 Coriolis force

1) A particle initially at rest, lies in a uniform zonal pressure gradient (along X). Assume the fluid unforced and the motion non dissipative, a planar motion and a uniform Coriolis parameter. With the help of the momentum equations, describe qualitatively the particle motion (assume a weak relative acceleration compared to the Coriolis component).

2) Consider a particle with mass M on an inclined plane (angle α), rotating around a vertical axis with angular velocity f(see figure).

This particle is initially at rest and we neglect friction and the centrifugal force.
Describe the particle motion.

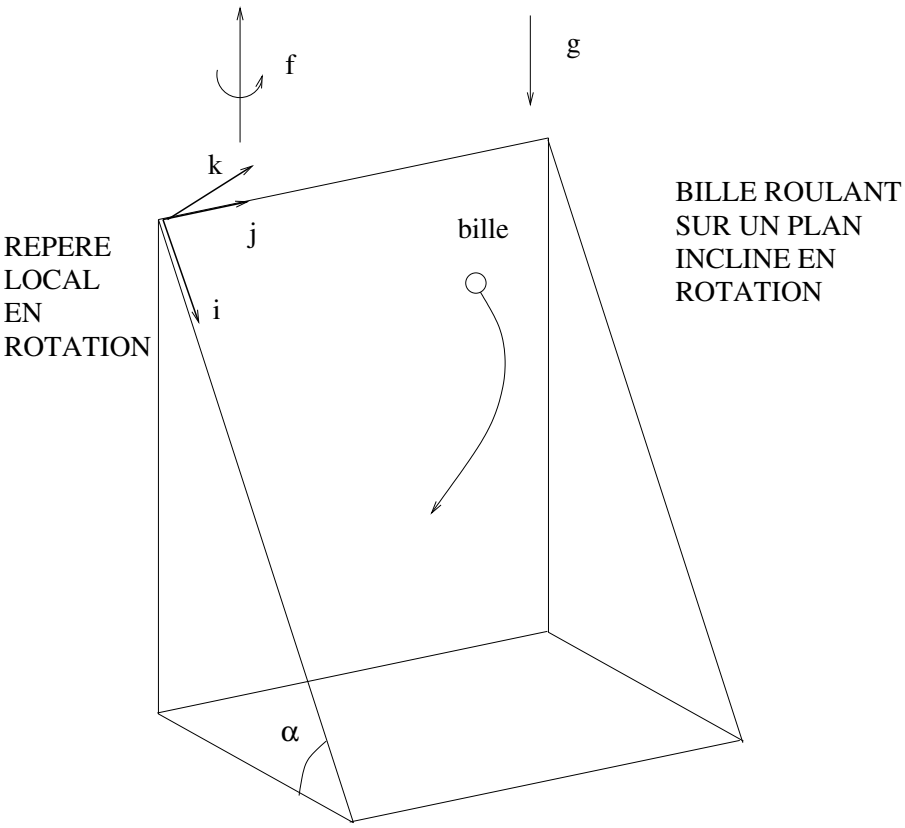


FIGURE 1 – Particle on the sloping plane